BRANCH	CY
SESSION	2024-25

SAGAR INSTITUTE OF SCIENCE & TECHNOLOGY(SISTec)
DEPARTMENT OF CSE- CYBER SECURITY

QUESTION BANK

NAME OF THE FACULTY: PROF. ABDULLAH DAR
SUBJECT/CODE : COMPUTER ORGANIZATION & ARCHITECTURE(CY-603)

UNIT-1

Q No.	QUESTIONS	Bloom's Taxonomy Level	Course Outcome
1	Explain the structure of a desktop computer with a neat block diagram and describe the function of each major unit.	1(Remembering)	CO1
2	Define Program Counter, Instruction Register, Memory Register, Control Word, and Stack Organization with suitable examples.	1(Remembering)	CO1
3	Explain the general register organization of CPU and describe the role of ALU in instruction execution.	2(Understanding)	CO1
4	Describe the instruction format and various addressing modes used in computer systems with examples.	2(Understanding)	CO1
5	Illustrate the concept of Register Transfer Language (RTL) and demonstrate bus and memory transfer operations using suitable	3(Applying)	CO1
6	Demonstrate the fetch and execution cycle of an instruction with the help of a flow diagram.	3(Applying)	CO1
7	Analyze the bus structure of a computer system and examine how CPU and memory communicate through system buses.	4(Analyzing)	CO1
8	Compare hardwired control unit and micro-programmed control unit with respect to design complexity, speed, and flexibility.	4(Analyzing)	CO1
9	Evaluate the role of microprogram sequencer and control memory in sequencing and execution of microinstructions.	5(Evaluating)	CO1
10	Design a functional architecture of a control unit incorporating microinstruction format and control memory organization.	6(Creating)	CO1

UNIT-2

Q No.	QUESTIONS	Bloom's Taxonomy Level	Course Outcome
1	Define 1's complement and 2's complement representation and list their advantages in signed arithmetic operations.	1(Remembering)	CO2
2	State the algorithmic steps involved in Booth's multiplication algorithm.	1(Remembering)	CO2
3	Explain signed addition and subtraction using 2's complement method with suitable examples.	2(Understanding)	CO2
4	Describe the procedure of floating-point arithmetic operations in computer systems.	2(Understanding)	CO2
5	Apply Booth's Algorithm to multiply two signed binary numbers.	3(Applying)	CO2
6	Perform binary division using restoring and non-restoring division methods with a suitable example	3(Applying)	CO2
7	Analyze the difference between fixed-point and floating-point number representation in terms of range and precision.	4(Analyzing)	CO2
8	Examine the hardware implementation requirements for arithmetic unit design.	4(Analyzing)	CO2
9	Evaluate the efficiency of Booth's algorithm compared to conventional multiplication methods	5(Evaluating)	CO2
10	Design an arithmetic unit capable of performing addition, subtraction, multiplication, and division operations.	6(Creating)	CO2